

Mixed partition functions are exactly the graph parameters of exponentially bounded edge-connection rank

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Abstract

We prove a conjecture of Regts and Sevenster: a complex-valued graph parameter f with $f(\emptyset) = 1$ has exponentially bounded edge-connection rank if and only if it is a mixed partition function; moreover, the model may be chosen with its numbers of even and odd colours explicitly bounded in terms of the rank bound. From f we construct a connection category, a rigid symmetric $\mathbb{C}$ -linear monoidal category whose morphism spaces have the connection ranks as dimensions and whose trace pairings are nondegenerate. The rank hypothesis forces moderate tensor growth, and a recent theorem of Etingof and Penneys then shows that every nilpotent endomorphism has trace zero; together with the nondegeneracy of the trace pairing, this makes the category semisimple, and a theorem of Deligne provides a faithful symmetric tensor functor to finite-dimensional super vector spaces. We then identify the resulting super tensor network with the Regts–Sevenster model exactly, viz. with its Eulerian-subgraph expansion and its sign of -1 for every fermionic circuit. An appendix gives an independent and direct proof of the nilpotent-trace step, showing that in a rigid symmetric $\mathbb{C}$ -linear category with $\text{End}(\mathbf{1}) = \mathbb{C}$, exponentially bounded endomorphism growth makes the trace zeta function of every endomorphism rational, with explicit degree bounds.

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1 Introduction

Which graph parameters are partition functions? The question has organised a substantial body of work since de la Harpe and Jones [7] introduced spin and vertex models on graphs as generalisations of the models of statistical mechanics — in later terminology, vertex-colouring and edge-colouring models respectively. Freedman, Lovász and Schrijver [14] characterised the partition functions of vertex-colouring models by reflection positivity and a rank condition on connection matrices; Szegedy [34] did the same for edge-colouring models, with the complex case characterised in [11]; and Schrijver characterised the partition functions of spin models [31] and of complex edge-colouring models [32] by rank growth. The connection-matrix method is treated at length in Lovász’s book [21], and its categorical undercurrent — that connection matrices are Gram matrices of a category of fragments — goes back at least to Lovász and Schrijver [22].

Regts and Sevenster [29], building on their symplectic models [27], enlarged the edge-colouring world by a fermionic sector: their *mixed partition functions*, recalled in Section 2.4, are partition functions of edge-colouring models on a super vector space $\mathbb{C}^{k|2\ell}$, and they include, for instance, every evaluation of the characteristic polynomial [29, Prop. 10]; the evaluation at zero, in particular, is not the partition function of any ordinary edge-colouring model [29, Prop. 9]. They proved that every mixed partition function has exponentially bounded edge-connection rank [29, Thm 6] and conjectured the converse. The history of that conjecture calls for care: a 2017 extended abstract announced the characterisation as a theorem [28, Thm 3.1], and the full paper records that the announcement was mistaken — the orthosymplectic Lie superalgebra had incorrectly been claimed to play no role — and states the converse as a conjecture instead [29, abstract and footnotes 1–2],

noting that the expected proof route through the invariant theory of the orthosymplectic supergroup meets a serious obstruction, the non-reductivity of $\mathfrak{osp}$; the completion of that route was deferred to a companion manuscript which has, to our knowledge, not appeared. A related conjecture — an algebraic characterisation of the mixed partition functions of a fixed $\mathbb{C}^{k|2\ell}$ — appears as Conjecture 5.7 of Sevenster’s thesis [33].

This paper proves the conjecture. The slogan is that a purely combinatorial exponential-rank condition forces a hidden finite-dimensional super tensor model; the precise statement is as follows.

Theorem 1.1 (Main Theorem). *A complex-valued graph parameter f , defined on finite multigraphs with free circles permitted and normalised by $f(\emptyset) = 1$, has exponentially bounded edge-connection rank if and only if it is a mixed partition function. More precisely, if $\mathrm{rk} M_{f,t} \leq R^t$ for some integer $R \geq 1$ and all $t \geq 0$, then $f = p_h$ for a functional $h \in (\mathrm{Sym} \mathbb{C}^k \otimes \Lambda \mathbb{C}^{2\ell})^*$ with $k, 2\ell \leq \lfloor 2eR \rfloor$, where e is the base of the natural logarithm.*

The implication from mixed partition functions to exponentially bounded rank is Regts and Sevenster’s theorem, with $R = \max(1, k + 2\ell)$; our contribution is the converse, together with the dimension bound. Note that the hypothesis includes $t = 0$, and this is essential: multiplicativity of f over disjoint unions is not assumed but *derived* from it (Lemma 3.2), a small but pleasant surprise.

The proof runs as follows. From f we construct a *connection category* $\mathcal{C}_f$: fragments modulo the kernel of the connection pairing, organised into a rigid symmetric monoidal category whose morphism spaces have the connection ranks as their dimensions and whose trace pairings are nondegenerate (Section 3). Its Cauchy completion $\mathcal{D}_f$ then satisfies the hypotheses of a recent theorem of Etingof and Penneys [13] — the rank bound becomes moderate tensor growth — so every nilpotent endomorphism has trace zero; combined with the nondegeneracy of the trace pairing this makes $\mathcal{D}_f$ semisimple abelian, and Deligne’s theorem [10] provides an exact faithful symmetric tensor functor to finite-dimensional super vector spaces (Section 4).

The remainder of the proof is the concrete half. The fibre functor yields a form, a copairing and one vertex tensor per degree, and f becomes the value of a super tensor network (Section 5). A separate reconstruction theorem then identifies that network, after a parity normalisation, with the Regts–Sevenster sum exactly — Eulerian subgraphs, orientations, pairings, and the sign of -1 for every fermionic circuit (Section 6).

The categorical route never touches the invariant theory of $\mathfrak{osp}$, which is how the non-reductivity obstruction is bypassed.

Two of the ingredients deserve explicit billing. First, the theorem of Etingof and Penneys, proved independently of the present work and in much greater generality, is what carries the semisimplicity step; Appendix A gives an independent and direct proof of that step for our setting, with a stronger conclusion — in a rigid symmetric $\mathbb{C}$ -linear category with $\mathrm{End}(\mathbf{1}) = \mathbb{C}$ and exponentially bounded endomorphism growth, the trace zeta function of *every* endomorphism is rational, with explicit degree bounds — and it is this quantitative form that yields the bound $k, 2\ell \leq \lfloor 2eR \rfloor$. Secondly, the reconstruction theorem of Section 6 is where the conjecture-specific work happens: Deligne’s theorem produces an abstract fibre functor, and turning that into the exact normalisations of [29, Definition 5] is a sign-sensitive computation that neither his theorem nor [13] performs. In adjacent work on the Holant framework, Young [36] has proved a converse Holant theorem deriving an orthogonal transformation from indistinguishability, and Cai and Young [6] analogous results through orbit closures for the general linear group (see also [5]); these concern a different notion of equivalence and do not characterise by rank growth.

The proof gives the coarse bound $k, 2\ell \leq \lfloor 2eR \rfloor$ on the colour dimensions, but it provides no algorithm for constructing the functional h and no effective bounds on its coordinates, Deligne’s functor being non-constructive; we do not pursue hypotheses weaker than a uniform exponential connection-rank bound; and we work in characteristic zero throughout.

Both implications of the Main Theorem, including the bound $k, 2\ell \leq \lfloor 2eR \rfloor$, have been machine-checked in Lean 4 over mathlib, with Deligne’s theorem [10, Thm 0.6] as the only axiomatised input [35]; Theorem A.1 is formalised in its stated generality, and the nilpotent-trace step is carried by it, not by [13]. The formal development presents the vertex functional in sorted colour order, in the sense of the convention discussion of Section 5.4.

1.1 Reading guide

The sections follow the proof architecture above literally: conventions and Definition 5 (§2), the connection category and its Cauchy completion (§3), semisimplicity and the fibre functor (§4), extraction (§5), reconstruction and the proof of the Main Theorem (§6), consequences (§7), and the independent proof of the nilpotent-trace step (Appendix A). The connection-category and reconstruction arguments are developed self-containedly; beyond them the proof uses stated results of Etingof–Penneys and of Deligne, and the standard representation theory of symmetric groups.

2 Mixed partition functions, fragments, and half-edge conventions

The purpose of this section is twofold: to fix our conventions for multigraphs, fragments and gluing, and to state the Regts–Sevenster definition of a mixed partition function together with the hypothesis class of the Main Theorem.

2.1 Graphs, half-edges, and parameters

A *graph* is a finite multigraph: loops, parallel edges, and *free circles* — closed edges incident with no vertex, isomorphic to the circle $\bigcirc$ — are all permitted. We write $\emptyset$ for the empty graph and $G \sqcup H$ for disjoint union. A *graph parameter* is a function from isomorphism classes of graphs to $\mathbb{C}$. We write $\mathbb{N} = \mathbb{Z}_{\geq 0}$.

Every edge other than a free circle consists of two *half-edges*, one at each endpoint; the two half-edges of a loop are distinct and lie at the same vertex. The degree $\deg(v)$ counts half-edges, so that a loop contributes two, and for $F \subseteq E$ we write $\delta_F(v)$ for the set of half-edges at v belonging to edges of F . Throughout this paper, orders, orientations and pairings of edges are carried by the half-edges, while colours are assigned to edges and inherited by both half-edges; this convention does real work at loops and parallel edges, and we shall rely on it without further comment. A fully formal treatment would encode a graph by a finite vertex set, a finite half-edge set, an incidence map, a perfect matching of half-edges into edges, and a nonnegative integer recording the free-circle components, which carry no half-edges; we shall not need this formal encoding explicitly.

2.2 Fragments and gluing

Intuitively, a fragment is a graph with some dangling edge-ends, which may later be glued to the dangling ends of another fragment. Formally, a *t-fragment* is a graph together with

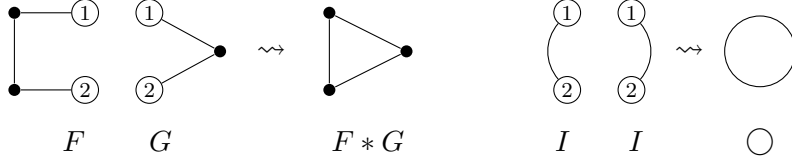


Figure 1: Gluing two fragments; the free-circle-creating case.

exactly t distinguished vertices of degree one, labelled $1, \dots, t$; the edge at the i -th labelled vertex is the *open end* i . We write $\mathfrak{F}_t$ for the set of isomorphism classes of t -fragments, so that $\mathfrak{F}_0$ is the set of graphs. To *glue* two open ends, delete the two labelled vertices and unify their edges into a single edge; for $F, G \in \mathfrak{F}_t$, the *full gluing* $F * G \in \mathfrak{F}_0$ glues the i -th open end of F to the i -th open end of G for every i . Note that gluing two open ends whose edges have both endpoints labelled produces a free circle; this is the reason free circles are admitted from the outset. We show two examples, including this degenerate case, in Figure 1.

Two further operations on fragments, composition and the identity strand, are introduced at the start of Section 3, where their elementary properties are established.

2.3 Connection matrices and the hypothesis class

The *edge-connection matrix* $M_{f,t}$ of a graph parameter f is the $\mathfrak{F}_t \times \mathfrak{F}_t$ matrix with entries

$$M_{f,t}(F, G) := f(F * G),$$

and its rank $\text{rk } M_{f,t}$ is the supremum of the ranks of its finite submatrices. The hypotheses of the Main Theorem are:

(H1) $f(\emptyset) = 1$;

(H2) there is an integer $R \geq 1$ with $\text{rk } M_{f,t} \leq R^t$ for all $t \geq 0$.

A real bound r may be replaced by $\lceil r \rceil$, so the integrality of R costs no generality. Note that $t = 0$ is included in (H2); this is essential, and is the source of the automatic multiplicativity of f (Lemma 3.2).

2.4 Mixed partition functions

The definition of a mixed partition function recalled in this subsection is Definition 5 of [29], and we cite it by that number, theirs, throughout the paper.

Fix $k, \ell \in \mathbb{N}$ and let $V = V_0 \oplus V_1$ be a super vector space with $\dim V_0 = k$ and $\dim V_1 = 2\ell$, with distinguished bases $e_1, \dots, e_k$ of V_0 and $\xi_1, \dots, \xi_{2\ell}$ of V_1 . To avoid confusion we write ξ_i, η_i where [29] write f_i, g_i , reserving the letter f for graph parameters throughout this paper; here

$$\eta_i := \begin{cases} -\xi_{i+\ell} & i \leq \ell, \\ \xi_{i-\ell} & i > \ell, \end{cases}$$

so that each η_i is a signed basis vector. Let $h \in (\text{Sym } V_0 \otimes \Lambda V_1)^*$. Regts and Sevenster permit an arbitrary such functional; since the exterior degree appearing at a vertex in the sum below is $\deg_F(v)$, which is even whenever F is Eulerian, the odd exterior part of h never contributes, and we may therefore impose, without loss of generality, that h vanishes on every component of odd exterior degree.

Call $F \subseteq E(G)$ *Eulerian* if $\deg_F(v)$ is even for every vertex v . For Eulerian F , choose an Eulerian orientation of F together with a compatible *local pairing* $\kappa = (\kappa_v)_v$, which pairs at each vertex v each incoming half-edge of $\delta_F(v)$ with an outgoing one; we list each pair as (a_1, a_2) with a_1 incoming. Following the pairs from edge to edge decomposes F into directed κ -circuits, and we write $c(\kappa)$ for their number. The *mixed partition function* of h is the graph parameter

$$p_h(G) := \sum_{\substack{F \subseteq E(G) \\ F \text{ Eulerian}}} (-1)^{c(\kappa)} \sum_{\psi: E \setminus F \rightarrow [k]} \sum_{\varphi: F \rightarrow [2\ell]} \prod_{v \in V(G)} h \left(\bigodot_{a \in \delta_{E \setminus F}(v)} e_{\psi(a)} \otimes \bigwedge_{(a_1, a_2) \in \kappa_v} \xi_{\varphi(a_1)} \wedge \eta_{\varphi(a_2)} \right),$$

for a graph G without free circles, extended to all graphs by $p_h(\bigcirc) := k - 2\ell$ and multiplicativity over disjoint union. Here ψ and φ colour edges, and each half-edge inherits the colour of its edge. The wedge over the pairs of κ_v is unambiguous: each factor $\xi_{\varphi(a_1)} \wedge \eta_{\varphi(a_2)}$ has even exterior degree, so the order of the pairs is immaterial and the displayed argument of h is a well-defined element of $\text{Sym } V_0 \otimes \Lambda V_1$. The value is independent of the choice of orientation and compatible local pairing: [29, Proposition 3] proves this for the skew sector, and the mixed sum reduces to it, one colouring ψ at a time, in the discussion preceding their Definition 5. A graph parameter is a *mixed partition function* if it equals p_h for some k, ℓ and h as above.

Regts and Sevenster proved that every mixed partition function satisfies (H2), with $R = \max(1, k + 2\ell)$ [29, Theorem 6] — the maximum matters only for the degenerate model with $k = 2\ell = 0$ — and (H1) is immediate, the empty graph carrying the empty product; the Main Theorem is the converse.

The definition is somewhat intricate, and so we now give an explicit example, to which we shall return. Take $k = 2$ and $\ell = 1$, fix $\theta \in \mathbb{C}$, and let $h = h(\theta)$ be determined by

$$h(e_1^{\odot i}) = \theta, \quad h(e_1^{\odot i} \odot e_2) = \sqrt{-1}, \quad h(e_1^{\odot i} \otimes \xi_1 \wedge \eta_1) = 1 \quad (i \geq 0),$$

and by $h = 0$ on all basis elements outside the span of these. Regts and Sevenster showed [29, Proposition 10] that, for graphs without free-circle components, $p_{h(\theta)}$ is the characteristic polynomial $G \mapsto \det(\theta I - A_G)$, where the adjacency matrix A_G counts a loop twice on the diagonal; since $k - 2\ell = 0$ here, $p_{h(\theta)}$ vanishes on every graph that does contain a free circle.

We evaluate $p_{h(\theta)}$ on the graph L with one vertex and one loop; see Figure 2. The loop has two half-edges at the single vertex v , so $\deg(v) = 2$, and both $F = \emptyset$ and $F = \{\text{loop}\}$ are Eulerian. For $F = \emptyset$, the vertex factor is $h(e_\psi \odot e_\psi)$: the colouring $\psi = 1$ contributes $h(e_1^{\odot 2}) = \theta$, while $\psi = 2$ contributes $h(e_2^{\odot 2}) = 0$. For $F = \{\text{loop}\}$, the pairing κ_v joins the loop's two half-edges into a single κ -circuit, so $c(\kappa) = 1$; the colouring $\varphi = j$ contributes $h(\xi_j \wedge \eta_j)$, and since

$$\xi_1 \wedge \eta_1 = -\xi_1 \wedge \xi_2 = \xi_2 \wedge \eta_2,$$

both odd colours contribute through the same basis vector of $\Lambda^2 V_1$, each with value 1. In total

$$p_{h(\theta)}(L) = \theta + (-1)^1 \cdot (1 + 1) = \theta - 2 = \det(\theta I - A_L),$$

as required, since $A_L = (2)$. It is worth pausing on what this small computation has already exercised: the two incidences of a loop, the Eulerian condition, the circuit sign, the distinction between a loop and a free circle, and the fact that the η -convention makes distinct odd colourings contribute through a common basis vector. All five phenomena return in earnest in Sections 5 and 6.



Figure 2: The loop graph L , its two Eulerian subsets, and the pairing κ .

3 The connection category and its Cauchy completion

We now organise fragments and gluing into a symmetric monoidal category $\mathcal{C}_f$, whose morphism spaces have the connection ranks as their dimensions, and complete it to a category $\mathcal{D}_f$ in which the remainder of the proof takes place. The category theory involved is elementary and self-contained: every trace identity below is proved by exhibiting an equality of glued graphs, and f is applied only afterwards. For readers of a combinatorial bent, the following dictionary may serve as a guide.

number of open ends	object
fragment	morphism
gluing	composition
disjoint union	tensor product
full closure	trace
connection pairing	trace pairing

We call $\mathcal{C}_f$ the *connection category* of f . To avoid confusion we do not use the term skein category, which is established in quantum topology for a related but distinct circle of ideas.

3.1 Composition of fragments

Fragments compose by partial gluing, and we fix the categorical order once and for all: a fragment $F \in \mathfrak{F}_{s+t}$ is read as a morphism from its first s open ends (its *inputs*) to its last t (its *outputs*), and for $F \in \mathfrak{F}_{s+t}$ and $G \in \mathfrak{F}_{t+u}$ the *composite* $G \circ F \in \mathfrak{F}_{s+u}$ — first F , then G , written right to left — glues the last t open ends of F to the first t open ends of G , in order. The inputs of $G \circ F$ are the inputs of F with their labels, and its outputs are the outputs of G , relabelled $s+1, \dots, s+u$. The *strand* $I \in \mathfrak{F}_2$ is the fragment with a single edge joining its two labelled vertices, and more generally, for a permutation π of $[t]$, the *permutation fragment* $P_\pi \in \mathfrak{F}_{t+t}$ consists of t disjoint strands joining open end i to open end $t+\pi(i)$. Whenever a disjoint union of fragments is read as a morphism, it carries the input–input–output–output block relabelling made explicit in Section 3.2.

Lemma 3.1. *The following hold for all fragments of compatible arities.*

- (i) *Composition is associative:* $(H \circ G) \circ F = H \circ (G \circ F)$.
- (ii) *The strand is an identity:* writing I^t for t disjoint strands, $F \circ I^s = F = I^u \circ F$ for $F \in \mathfrak{F}_{s+u}$.
- (iii) *Composition and disjoint union interchange:* for $F \in \mathfrak{F}_{s+t}$, $G \in \mathfrak{F}_{t+u}$, $F' \in \mathfrak{F}_{s'+t'}$ and $G' \in \mathfrak{F}_{t'+u'}$, $(G \sqcup G') \circ (F \sqcup F') = (G \circ F) \sqcup (G' \circ F')$.

- (iv) *Gluing commutes with simultaneous relabelling of the interface, and $P_\pi \circ P_\sigma = P_{\pi\sigma}$, where $\pi\sigma$ denotes the composite function (σ first).*

Proof. All four statements are instances of one description of iterated gluing. A gluing is the quotient of the edge multiset of a disjoint union along a pairing of open ends: each matched pair of pendant edges is unified into one edge, chains of matched strands collapse into single edges, and closed chains collapse into free circles. In an iterated composite each open end participates in at most one interface pair, so the union of the interface pairings is itself a single pairing on ends, and unifying iteratively in any order equals unifying simultaneously along that union; this proves (i). For (ii), gluing a strand's end to an open end o of F deletes the two labelled vertices and unifies the strand's edge with the edge of o , yielding a single pendant edge ending at the strand's other labelled vertex, which is F with that open end relabelled. For (iii), the interface pairing of the left-hand side splits as the disjoint union of the blockwise pairings, and quotienting a disjoint union along a split pairing is the disjoint union of the quotients. Finally (iv): a relabelling is an isomorphism of the data the pairing is defined on, and composing permutation fragments concatenates strands, which by (ii) collapse to the strands of the composite permutation. $\square$

3.2 The connection category

For the remainder of the paper, f denotes a graph parameter satisfying (H1) and (H2) with rank bound R . We begin with a small surprise: multiplicativity of f need not be assumed, because the $t = 0$ case of (H2) provides it.

Lemma 3.2. *For all graphs G and H we have $f(G \sqcup H) = f(G)f(H)$.*

Proof. At $t = 0$, hypothesis (H2) reads $\text{rk } M_{f,0} \leq R^0 = 1$, and $M_{f,0}(G, H) = f(G \sqcup H)$. The row of $\emptyset$ is $(f(H))_H$, which is nonzero since $f(\emptyset) = 1$ by (H1). A matrix of rank at most one with a nonzero row has every row a scalar multiple of it: $f(G \sqcup H) = c_G f(H)$ for some $c_G \in \mathbb{C}$, and setting $H = \emptyset$ gives $c_G = f(G)$. $\square$

Let $\mathbb{C}^{(\mathfrak{F}_t)}$ denote the vector space of finite formal $\mathbb{C}$ -combinations of t -fragments, and let

$$\Theta_t: \mathbb{C}^{(\mathfrak{F}_t)} \rightarrow (\mathbb{C}^{(\mathfrak{F}_t)})^*, \quad \Theta_t(F) := (G \mapsto f(F * G)),$$

extended linearly, with kernel $N_t := \ker \Theta_t$. The quotient $\mathbb{C}^{(\mathfrak{F}_t)}/N_t$ has dimension exactly $\text{rk } M_{f,t}$: it is isomorphic to the image of Θ_t , i.e. to the row space of the connection matrix, and any finite linearly independent family of rows remains independent after restriction to a suitable finite set of columns, so the row-space dimension equals the supremum of the finite-submatrix ranks. By (H2) this dimension is at most R^t ; this is how the rank hypothesis will enter every dimension count below.

Lemma 3.3. *The kernels N form an ideal for both operations, in the following sense.*

- (a) *If $x \in N_{s+t}$ and w, y are fragments of compatible arities, then $w \circ x \circ y \in N$.*
- (b) *If $x \in N_{s+t}$ and $z \in \mathfrak{F}_{s'+t'}$, then $x \sqcup z \in N_{(s+s')+(t+t')}$.*

Proof. For (a), any full closure of $w \circ x \circ y$ against a fragment G performs, by Lemma 3.1(i), the same simultaneous unification as the full closure of x against $y \circ G \circ w$; since $x \in N$, the value of f on the latter vanishes for every G . For (b), given a closure G of $x \sqcup z$, first glue the open ends of z into G , obtaining a fragment G_z of arity $s+t$, possibly with extra closed components; then $f((x \sqcup z) * G) = f(x * G_z) = 0$. $\square$

We may now define the category. The *connection category* $\mathcal{C}_f$ has objects the natural numbers, the object t being written $X^{\otimes t}$; its morphism spaces are

$$\mathrm{Hom}(X^{\otimes s}, X^{\otimes t}) := \mathbb{C}^{\mathfrak{F}_{s+t}} / N_{s+t},$$

a fragment being read as a morphism from its first s open ends to its last t ; composition is induced by the composition of fragments, which descends to the quotients by Lemma 3.3(a), and the identity of $X^{\otimes t}$ is the class of I^t , by Lemma 3.1(ii). The tensor product is addition on objects and is induced by disjoint union on morphisms, with the block relabelling that the input–output reading demands: for $F \in \mathfrak{F}_{s+t}$ and $F' \in \mathfrak{F}_{s'+t'}$, the disjoint union is read as a morphism with inputs the inputs of F followed by those of F' and outputs the outputs of F followed by those of F' ; concretely, F -input $i \mapsto i$, F' -input $i \mapsto s + i$, F -output $j \mapsto s + s' + j$, and F' -output $j \mapsto s + s' + t + j$. This descends by Lemma 3.3(b); it is strictly associative and unital with unit $\mathbf{1} := X^{\otimes 0}$, since disjoint union of isomorphism classes is so. Thus $\mathcal{C}_f$ is a $\mathbb{C}$ -linear strict monoidal category.

Lemma 3.4. $\mathrm{End}(\mathbf{1}) = \mathbb{C} \cdot [\emptyset]$.

Proof. By construction $\dim \mathrm{End}(\mathbf{1}) = \mathrm{rk} M_{f,0}$, which is 1 by (H1) and (H2), and $[\emptyset] \neq 0$ since $f(\emptyset * \emptyset) = 1$. $\square$

We henceforth identify $\mathrm{End}(\mathbf{1})$ with $\mathbb{C}$ via $[\emptyset] \mapsto 1$; every closed graph G thereby represents the scalar $f(G)$. Indeed $[G] = f(G)[\emptyset]$: for every closed graph H , $f((G - f(G)\emptyset) \sqcup H) = f(G \sqcup H) - f(G)f(H) = 0$ by Lemma 3.2, so $G - f(G)\emptyset \in N_0$.

The category is symmetric. The braiding of X with itself is the class of $P_{(12)}$, and more generally the classes of the permutation fragments realise every permutation of tensor factors; by Lemma 3.1(iv) they compose as the permutations do. Since the monoidal structure is strict, the hexagon axioms reduce to identities among permutation fragments of the form $P_{\pi \sqcup \mathrm{id}} \circ P_{\mathrm{id} \sqcup \pi}$, which hold by the same composition rule; naturality of the braiding is Lemma 3.1(iv) applied to the relabelling the braiding performs; and the braiding squares to the identity.

The object X is self-dual. Both the coevaluation $\mathrm{coev} \in \mathrm{Hom}(\mathbf{1}, X^{\otimes 2})$ and the evaluation $\mathrm{ev} \in \mathrm{Hom}(X^{\otimes 2}, \mathbf{1})$ are the class of the strand I , read with both open ends as outputs and both as inputs respectively; the snake identities are literal strand unifications, viz. instances of Lemma 3.1(ii). Consequently every object of $\mathcal{C}_f$ is self-dual: the evaluation and coevaluation of $X^{\otimes t}$ are the classes of t parallel strands, the i -th end of one block joined to the i -th end of the other, and the snake identities again reduce to strand unifications. The *trace* of $u \in \mathrm{End}(X^{\otimes t})$ is its full strand closure — the scalar obtained by gluing the i -th output back to the i -th input for every i . Every trace is thus an f -value of a closed graph. In particular

$$\mathrm{tr}(\mathrm{id}_X) = f(\bigcirc),$$

the categorical dimension of X : closing a single strand produces the free circle, which is the reason free circles were admitted in Section 2.1.

Finally, closed graphs decompose into stars. Let $\sigma_d \in \mathrm{Hom}(\mathbf{1}, X^{\otimes d})$ denote the class of the d -star: a single vertex with d pendant edges, its open ends labelled $1, \dots, d$; in particular σ_0 is the isolated vertex. For a closed graph G without free circles, fix orders on its vertices and edges, and for each edge an order of its two half-edges; then in $\mathcal{C}_f$

$$[G] = \left(\bigotimes_{e \in E} \mathrm{ev} \right) \circ P_G \circ \left(\bigotimes_{v \in V} \sigma_{\deg v} \right), \quad (1)$$

where P_G is the permutation fragment matching each star leg to its half-edge. Indeed, the right-hand side glues the stars' pendant edges in pairs along E , by Lemma 3.1(i) and (iii), and each matched pair unifies into the single edge e , reconstructing G ; a loop sends two legs of the same star to one evaluation, which is where the half-edge convention of Section 2.1 earns its keep.

3.3 The trace calculus

The following identities are standard consequences of rigidity and symmetry, but in $\mathcal{C}_f$ they admit direct combinatorial proofs: each asserts that two closed graphs are equal, and f is applied only afterwards. We prove them in this form because Section 4 will lean on them before any further structure is available. We abbreviate $uv := u \circ v$.

Lemma 3.5. *Let u and v be morphisms of $\mathcal{C}_f$ of appropriate types.*

- (a) $\text{tr}(uv) = \text{tr}(vu)$ for $u: Y \rightarrow Z$ and $v: Z \rightarrow Y$.
- (b) $\text{tr}_{Y \otimes Z}(u \otimes v) = \text{tr}(u) \text{tr}(v)$ for $u \in \text{End}(Y)$, $v \in \text{End}(Z)$.
- (c) $\text{tr}(c_{Z,Z} \circ (u \otimes v)) = \text{tr}(uv)$ for $u, v \in \text{End}(Z)$, where c is the braiding.

Proof. (a) Both traces are the f -value of the same closed graph: the strand closure of $u \circ v$ glues every open end of u to the matching open end of v — the composition interface directly, and the outer legs through the closing strands, which merge away by Lemma 3.1(ii) — and the closure of $v \circ u$ performs the identical set of end-pairings. (b) The closure of a disjoint union is the disjoint union of the closures, and f is multiplicative by Lemma 3.2. (c) Closing $c \circ (u \otimes v)$ connects the output of u through the swap to the input of v , and the output of v back to the input of u ; the resulting closed graph is the cyclic gluing of u and v , which is exactly the closure of $u \circ v$, the interface strands again merging away. $\square$

Lemma 3.6. *If $u \in \text{Hom}(X^{\otimes s}, X^{\otimes t})$ satisfies $\text{tr}(uv) = 0$ for every $v \in \text{Hom}(X^{\otimes t}, X^{\otimes s})$, then $u = 0$.*

Proof. The trace closure of u against a fragment v differs from the defining pairing $u * G$ of Section 3.2 only by a fixed relabelling of the $s + t$ open ends. Relabelling is a bijection on fragments, so the two families of test values coincide, and the hypothesis places a lift of u in N_{s+t} . $\square$

In the language of tensor categories, Lemma 3.6 says that $\mathcal{C}_f$ has no nonzero negligible morphisms; it is the single point at which the definition of $\mathcal{C}_f$ as a quotient *by* the connection kernel pays off, and it is the engine of semisimplicity in Section 4.

3.4 The Cauchy completion

The category $\mathcal{C}_f$ has neither direct sums nor images of idempotents, and both will be needed. We therefore pass to the *Cauchy completion*

$$\mathcal{D}_f := \text{Kar}(\text{Add}(\mathcal{C}_f)),$$

where Add is the additive envelope — objects are finite tuples of objects of $\mathcal{C}_f$, morphisms are matrices — and Kar is the Karoubi envelope — objects are pairs (Y, e) with $e \in \text{End}(Y)$ idempotent, and morphisms $(Y, e) \rightarrow (Z, e')$ are the $a: Y \rightarrow Z$ with $a = e'ae$. The Karoubi

envelope of an additive category is additive, with $(Y, e) \oplus (Z, e') = (Y \oplus Z, e \oplus e')$, so $\mathcal{D}_f$ admits finite direct sums and split idempotents together.¹

The structure of $\mathcal{C}_f$ extends. The tensor product extends bilinearly to tuples and to idempotent pairs, with the symmetry acting componentwise; morphism spaces remain finite-dimensional, being submatrices of matrices of morphism spaces of $\mathcal{C}_f$; $\text{End}(\mathbf{1}) = \mathbb{C}$ persists, since $\mathbb{C}$ has no nontrivial idempotents; and duals extend by $(Y_1, \dots, Y_m)^\vee = (Y_1^\vee, \dots, Y_m^\vee)$ and $(Y, e)^\vee = (Y^\vee, e^\vee)$, with evaluation and coevaluation corrected by the idempotents — the cup is $(e \otimes e^\vee) \circ \text{coev}_Y$ and the cap $\text{ev}_Y \circ (e^\vee \otimes e)$; the first snake identity for (Y, e) then unwinds to the defining formula of the dual morphism $e^\vee$, and the second collapses onto the snake for Y , so no new computation arises, and on tuples the snakes reduce entrywise. Traces are computed accordingly: the trace of a matrix of endomorphisms is the sum of the traces of its diagonal entries, and the trace of $a \in \text{End}(Y, e)$ is $\text{tr}_Y(a)$. Thus $\mathcal{D}_f$ is a rigid symmetric $\mathbb{C}$ -linear monoidal category with finite-dimensional morphism spaces and $\text{End}(\mathbf{1}) = \mathbb{C}$, and it is generated by X under tensor products, direct sums and summands.

The nondegeneracy of Lemma 3.6 survives the completion; this small lemma is what will kill the Jacobson radical in Section 4, so we record it with care.

Lemma 3.7. *Let P and Q be objects of $\mathcal{D}_f$ and let $a: P \rightarrow Q$ satisfy $\text{tr}(ba) = 0$ for every $b: Q \rightarrow P$. Then $a = 0$.*

Proof. First let P and Q be objects of $\text{Add}(\mathcal{C}_f)$, say with components $(Y_i)_i$ and $(Z_j)_j$, and let $a = (a_{ji})$ as a matrix. For fixed indices i and j , let b be the matrix supported in entry (i, j) with value an arbitrary $b_{ij}: Z_j \rightarrow Y_i$. Then $\text{tr}(ba) = \text{tr}(b_{ij}a_{ji}) = 0$ for every b_{ij} , and Lemma 3.6 gives $a_{ji} = 0$; as i and j were arbitrary, $a = 0$.

Now let $P = (Y, e)$ and $Q = (Z, e')$ be objects of $\mathcal{D}_f$ and $a = e'ae: P \rightarrow Q$ with $\text{tr}(ba) = 0$ for all $b: Q \rightarrow P$. For an arbitrary $c: Z \rightarrow Y$ in $\text{Add}(\mathcal{C}_f)$, put $b := ece'$, a morphism $Q \rightarrow P$. Then, using $a = e'ae$ and Lemma 3.5(a),

$$\text{tr}(ca) = \text{tr}(ce'ae) = \text{tr}(ece'a) = \text{tr}(ba) = 0,$$

and the first paragraph gives $a = 0$. □

4 Moderate growth, nilpotent traces, and the super fibre functor

We are now ready to prove that $\mathcal{D}_f$ is semisimple abelian and admits a super fibre functor. The whole argument rests on three facts: every object of $\mathcal{D}_f$ has moderate tensor growth; nilpotent endomorphisms have trace zero; and the trace pairing is nondegenerate (Lemma 3.7). The rest is finite-dimensional algebra and a theorem of Deligne. Semisimplicity is *not* automatic for rigid symmetric categories — without a growth hypothesis there exist such categories containing nilpotent endomorphisms of nonzero trace (see [13, §3.4] for examples) — and this is precisely why the present section has content.

¹The order matters: applying Add *after* Kar need not produce an idempotent-complete category, since a summand of a direct sum need not be a direct sum of summands. The distinction disappears once $\mathcal{D}_f$ is known to be semisimple (Section 4), after which $\text{Add}(\text{Kar}(\mathcal{C}_f)) \simeq \mathcal{D}_f$.

4.1 Moderate growth and nilpotent traces

Lemma 4.1. *Let $Z \in \mathcal{D}_f$ be a direct summand of $\bigoplus_{j=1}^m X^{\otimes n_j}$ with $m \geq 1$, and put $n_{\max} := \max_j n_j$. Then*

$$\dim \text{End}(Z^{\otimes N}) \leq m^{2N} R^{2n_{\max}N} \quad (N \geq 0).$$

In particular $\dim \text{End}(Z^{\otimes N}) < N!$ for all sufficiently large N . Since every object of $\mathcal{D}_f$ is such a summand, every object of $\mathcal{D}_f$ has moderate growth in the sense of [13].

Proof. Put $Y := \bigoplus_{j=1}^m X^{\otimes n_j}$ and choose $\iota: Z \rightarrow Y$ and $\rho: Y \rightarrow Z$ with $\rho\iota = \text{id}_Z$. Tensoring this retraction shows that $Z^{\otimes N}$ is a retract of $Y^{\otimes N}$, and $g \mapsto \iota^{\otimes N} g \rho^{\otimes N}$ embeds $\text{End}(Z^{\otimes N})$ linearly into $\text{End}(Y^{\otimes N})$, with left inverse $a \mapsto \rho^{\otimes N} a \iota^{\otimes N}$. The N -th tensor power of the sum expands into m^N summands $X^{\otimes |w|}$ indexed by words w , each with $|w| \leq n_{\max}N$, so

$$\dim \text{End}(Z^{\otimes N}) \leq \sum_{w, w'} \dim \text{Hom}(X^{\otimes |w|}, X^{\otimes |w'|}) \leq m^{2N} R^{2n_{\max}N},$$

using (H2) for each summand. The right-hand side is exponential in N , hence eventually smaller than $N!$. $\square$

Lemma 4.2. *The quantum trace of [13] on $\mathcal{D}_f$ coincides with the closure trace of Section 3.3.*

Proof. Etingof and Penneys construct their quantum trace from the Drinfeld isomorphism [13, §2.3]. In a symmetric rigid category, for $u \in \text{End}(Y)$, that construction reduces to the composite

$$\mathbf{1} \xrightarrow{\text{coev}_Y} Y \otimes Y^\vee \xrightarrow{u \otimes \text{id}} Y \otimes Y^\vee \xrightarrow{c_{Y, Y^\vee}} Y^\vee \otimes Y \xrightarrow{\text{ev}_Y} \mathbf{1}.$$

In $\mathcal{C}_f$ the structural morphisms coev_Y , $c_{Y, Y^\vee}$ and ev_Y are represented by strand and permutation fragments, and inserting u into their composite performs exactly the end-pairings of the full strand closure of u , by Lemma 3.1. The identification extends additively to $\text{Add}(\mathcal{C}_f)$, and for a Karoubi object (Y, e) it restricts to the idempotent summand: a morphism $a = eae$ has trace $\text{tr}_Y(a)$ on both sides. $\square$

We shall apply the following theorem, extracted from Corollary 1.5 of [13] — which gives the vanishing for non-negligible indecomposable objects — together with their Lemma 1.6, which extends it to every object of a rigid category.

Theorem 4.3 (Etingof–Penneys [13, Cor. 1.5, Lem. 1.6]). *Let $\mathcal{B}$ be a Cauchy complete $\mathbb{C}$ -linear braided monoidal category with finite-dimensional morphism spaces and $\text{End}(\mathbf{1}) = \mathbb{C}$, which is rigid and in which every object has moderate growth. Then the quantum trace of every nilpotent endomorphism in $\mathcal{B}$ is zero.*

Theorem 4.4. *Every nilpotent endomorphism in $\mathcal{D}_f$ has trace zero.*

Proof. The category $\mathcal{D}_f$ satisfies every hypothesis of Theorem 4.3: it is Cauchy complete by construction (Section 3.4); it is $\mathbb{C}$ -linear and symmetric, hence braided; its morphism spaces are finite-dimensional; $\text{End}(\mathbf{1}) = \mathbb{C}$ (Lemma 3.4); it is rigid; and every object has moderate growth (Lemma 4.1). By Lemma 4.2 the quantum trace is our trace.² $\square$

Appendix A proves Theorem 4.4 independently of [13]: apply Theorem A.1 with $A = m^2 R^{2n_{\max}}$.

²Corollary 1.8 of [13] packages the remainder of this section's argument through semisimplification; we prefer the route below because it exposes the role of the connection pairing.

4.2 Semisimplicity

Theorem 4.5. *For every $Y \in \mathcal{D}_f$ the algebra $\text{End}(Y)$ is semisimple.*

Proof. Let j be an element of the Jacobson radical of the finite-dimensional algebra $\text{End}(Y)$ and let $u \in \text{End}(Y)$ be arbitrary. Then ju lies in the radical, hence is nilpotent — the radical of a finite-dimensional algebra is a nilpotent ideal — so $\text{tr}(ju) = 0$ by Theorem 4.4, and by cyclicity (Lemma 3.5(a)) also $\text{tr}(uj) = 0$. Now Lemma 3.7 applied with $a = j$ gives $j = 0$. $\square$

Decompose the identity of any $Y \in \mathcal{D}_f$ into orthogonal primitive idempotents of the semisimple algebra $\text{End}(Y)$; since idempotents split in $\mathcal{D}_f$, this expresses Y as a finite direct sum of objects with primitive identity. We call an object S with $\text{End}(S) = \mathbb{C}$ an *atom* — the word *simple* is reserved until the abelian structure is available — and the summands just produced are atoms, because a corner $e \text{End}(Y) e$ of a semisimple $\mathbb{C}$ -algebra at a primitive idempotent is a finite-dimensional division algebra over the algebraically closed field $\mathbb{C}$, hence $\mathbb{C}$ itself.

Lemma 4.6. *A nonzero morphism between atoms is an isomorphism, and non-isomorphic atoms admit no nonzero morphisms between them.*

Proof. Let $a: S \rightarrow T$ be nonzero with S, T atoms. By Lemma 3.7 there is $b: T \rightarrow S$ with $\text{tr}(ba) \neq 0$. Since $\text{End}(S) = \mathbb{C}$ we have $ba = \mu \text{id}_S$ with $\mu \neq 0$, so $v := \mu^{-1}b$ satisfies $va = \text{id}_S$; then $av \in \text{End}(T) = \mathbb{C}$ is an idempotent, nonzero because $ava = a \neq 0$, hence $av = \text{id}_T$. Thus a is an isomorphism. $\square$

Theorem 4.7. *Choose representatives $\{S_\gamma\}$ of the isomorphism classes of atoms. The functor $Y \mapsto (\text{Hom}(S_\gamma, Y))_\gamma$ is an equivalence of $\mathbb{C}$ -linear categories between $\mathcal{D}_f$ and the category of finite-support families of finite-dimensional vector spaces. Consequently $\mathcal{D}_f$ is semisimple abelian, its tensor product is exact in each variable, and its simple objects are the atoms.*

Proof. Every object is a finite direct sum of atoms, as shown above. By Lemma 4.6, for $Y \cong \bigoplus S_{\gamma(i)}$ and $Z \cong \bigoplus S_{\delta(j)}$ every morphism is a matrix over the spaces $\text{Hom}(S_\gamma, S_\delta) = \delta_{\gamma\delta} \cdot \mathbb{C}$, so the functor is fully faithful; a finite-support family (W_γ) is the image of $\bigoplus_\gamma S_\gamma^{\oplus \dim W_\gamma}$, so it is essentially surjective. The target category is semisimple abelian, and abelian-ness transports along the equivalence. Exactness of the tensor product holds because in a semisimple abelian category every short exact sequence splits and additive functors preserve split exactness. $\square$

4.3 The super fibre functor

We may now verify the hypotheses of Deligne's theorem for $\mathcal{D}_f$:

- essentially small: the fragments form a set of representatives, and all Hom-spaces, idempotents, tuples and pairs built from them are set-sized;
- abelian, $\mathbb{C}$ -linear and semisimple, with finite-dimensional morphism spaces and with $\text{End}(\mathbf{1}) = \mathbb{C}$: Theorem 4.7 and Section 3.4;
- rigid and symmetric, with exact tensor product: Section 3.4 and Theorem 4.7;

- finitely tensor-generated by the self-dual object X : every object of $\mathcal{D}_f$ is a direct summand of a finite direct sum of tensor powers of X ; summands are subobjects in the semisimple abelian structure, and X is self-dual, so this is tensor generation in Deligne's sense;
- every object of moderate growth: in a semisimple category the length of $Y^{\otimes N}$ is at most $\dim \text{End}(Y^{\otimes N})$ — if $Y^{\otimes N} \cong \bigoplus_i S_i^{\oplus m_i}$ then the length is $\sum_i m_i \leq \sum_i m_i^2 = \dim \text{End}$ — and the latter is exponentially bounded by Lemma 4.1.

In this list the nilpotent-trace theorem enters only through Theorem 4.7 — the semisimple abelian structure, together with the tensor exactness and the subobject language that ride on it — and everything else follows from the rank bound and the constructions of Section 3 alone. The representation-theoretic input thus enters the route to the fibre functor at exactly one point.

Theorem 4.8 (Deligne [10, Thm 0.6]; see also [26]). *There is an exact faithful $\mathbb{C}$ -linear symmetric monoidal functor $\omega: \mathcal{D}_f \rightarrow \text{sVect}_{\mathbb{C}}$.*

The remainder of the proof consumes less than the theorem provides: only the existence of a symmetric monoidal $\mathbb{C}$ -linear functor to $\text{sVect}_{\mathbb{C}}$ is used below — the nondegeneracy of b , the evenness and invariance of the vertex tensors, and the transfer of the square's death are all carried by functoriality alone, and the exactness and faithfulness of ω play no further role.

Here the morphisms of $\text{sVect}_{\mathbb{C}}$ are the parity-preserving linear maps; we use this convention repeatedly. We fix such an ω once and for all, suppress its structural isomorphisms $\omega(Y) \otimes \omega(Z) \cong \omega(Y \otimes Z)$ from the notation throughout, and write $V := \omega(X)$, a super vector space $V = V_0 \oplus V_1$ with $\dim V_0 = k$ and, as we shall see in Section 5, $\dim V_1 = 2\ell$ even.

The growth hypothesis also bounds the dimensions of V ; this is the source of the quantitative clause of the Main Theorem. We borrow from Appendix A exactly one statement, whose elementary symmetric-group proof is independent of everything in this section. Here Φ_n denotes the symmetric-braiding action of $\mathbb{C}[S_n]$ on the n -th tensor power, and e_λ the central idempotent of the block of $\lambda \vdash n$.

Lemma 4.9 (Square death; proved in Appendix A). *Let Z be an object of a rigid symmetric $\mathbb{C}$ -linear monoidal category with $\text{End}(\mathbf{1}) = \mathbb{C}$, satisfying $\dim \text{End}(Z^{\otimes n}) \leq A^n$ for all n and some constant $A \geq 1$, and let s be an integer with $s > 2e\sqrt{A}$. Then $\Phi_{s^2}(e_{(s^s)}) = 0$ in $\text{End}(Z^{\otimes s^2})$.*

Proof. This is Lemma A.5, whose proof combines the block-injection estimate of Lemma A.3 with a hook-length bound. $\square$

Corollary 4.10. *The dimensions of V satisfy $k \leq \lfloor 2eR \rfloor$ and $\dim V_1 \leq \lfloor 2eR \rfloor$.*

Proof. Apply Lemma 4.9 to $Z = X$ in $\mathcal{D}_f$, with growth constant $A = R^2$ by (H2): any integer $s > 2eR$ — in particular the least one, $s = \lfloor 2eR \rfloor + 1$ — has $\Phi_{s^2}(e_{(s^s)}) = 0$ in $\text{End}(X^{\otimes s^2})$. Here, for $\lambda \vdash n$, choose a primitive idempotent q_λ in the λ -block of $\mathbb{C}[S_n]$ and define $S_\lambda(X) := \text{im } \Phi_n(q_\lambda)$, an image that exists because idempotents split in $\mathcal{D}_f$. The block being $\text{Mat}_{d_\lambda}(\mathbb{C})$, the image of its central identity e_λ is a direct sum of d_λ isomorphic copies of $S_\lambda(X)$; so $\Phi_{s^2}(e_{(s^s)})$ projects onto the (s^s) -isotypic summand $S_{(s^s)}(X)^{\oplus d_{(s^s)}}$ of $X^{\otimes s^2}$, and its vanishing is equivalent to $S_{(s^s)}(X) = 0$ in $\mathcal{D}_f$. Applying the $\mathbb{C}$ -linear symmetric monoidal functor ω carries $\Phi_{s^2}(e_{(s^s)})$ to the corresponding idempotent

for the super-permutation action on $V^{\otimes s^2}$, so $S_{(s^s)}(V) = 0$ in $\mathbf{sVect}_{\mathbb{C}}$. By the hook criterion for super vector spaces [10, Cor. 1.9] (in substance the theorem of Berele and Regev [3]), $S_{\lambda}(V) \neq 0$ exactly when λ lies in the fat hook $H(\dim V_0, \dim V_1)$, where $H(a, b) := \{\lambda : \lambda_{a+1} \leq b\}$ (Appendix A); the square (s^s) lies outside a fat hook $H(a, b)$ precisely when $s > a$ and $s > b$. Hence $\dim V_0 < s$ and $\dim V_1 < s$, i.e. both are at most $\lfloor 2eR \rfloor$. Note that the square is its own conjugate partition, so the conclusion is insensitive to the row/column convention in the hook criterion. $\square$

5 From categorical tensors to a super tensor network

In this section we extract from the fibre functor the raw data of a mixed model — a bilinear form, a copairing, and one vertex tensor per degree — and express f as the value of a super tensor network built from them. This section and the next are where signs can go wrong; each sign below is paid exactly once, in a numbered statement. The data map is

$$\text{ev} \mapsto b, \quad \text{coev} \mapsto C, \quad \sigma_d \mapsto T_d \rightsquigarrow h_d^{\text{cat}}, \quad (C, h^{\text{cat}}) \rightsquigarrow (\widehat{C}, h^{\text{RS}}) \rightsquigarrow \text{Definition 5},$$

and the subsections follow it: the form and its coordinates (§5.1), the copairing and its twist (§5.2), the vertex tensors and the passage to a tensor network (§5.3), the change of normalisation (§5.4); the identification with Definition 5 is then the business of Section 6.

5.1 The form and its coordinates

Write $b := \omega(\text{ev}): V \otimes V \rightarrow \mathbb{C}$. The form b is even, since ω preserves the grading; supersymmetric, since $\text{ev} \circ c = \text{ev}$ holds already in $\mathcal{C}_f$ (gluing the swap into a strand yields a strand); and nondegenerate, since the snake identities make $x \mapsto b(x, -)$ invertible with inverse built from $\omega(\text{coev})$. The restriction of b to V_1 is then a nondegenerate antisymmetric form, so $\dim V_1$ is even; we write $\dim V_1 = 2\ell$, and Corollary 4.10 bounds both $k = \dim V_0$ and 2ℓ by $\lfloor 2eR \rfloor$. Choose coordinates adapted to b : a basis $e_1, \dots, e_k$ of V_0 with $b(e_i, e_j) = \delta_{ij}$ and $b(e_i, V_1) = 0$, and a basis $\xi_1, \dots, \xi_{2\ell}$ of V_1 with

$$b(\xi_m, \xi_{m+\ell}) = 1 = -b(\xi_{m+\ell}, \xi_m) \quad (m \leq \ell), \quad b(\xi_i, \xi_j) = 0 \text{ otherwise.}$$

With η_i as in Section 2.4, this gives $b(\xi_i, \eta_i) = -1$ and $b(\eta_i, \xi_j) = \delta_{ij}$ for all i, j ; the latter identity is verified entry by entry in Lemma 6.2 below.

For homogeneous $v_i, w_i \in V$ define the *tensor-power pairing*

$$\beta_d(v_1 \otimes \dots \otimes v_d, w_1 \otimes \dots \otimes w_d) := (-1)^{\sum_{i < j} |v_j| |w_i|} \prod_{i=1}^d b(v_i, w_i).$$

Lemma 5.1. *For a fixed word of parities, the Koszul sign of a rearrangement in $\mathbf{sVect}_{\mathbb{C}}$ equals the sign of the permutation it induces on the odd letters; in particular the sign is multiplicative under composition, and signed evaluations along any two rearrangement paths realising the same permutation of the labelled homogeneous tensor factors are equal. Consequently:*

- (a) β_d is the signed evaluation obtained by rearranging $(v_1 \dots v_d w_1 \dots w_d)$ into the interleaved word $(v_1 w_1 v_2 w_2 \dots v_d w_d)$ and applying b to each adjacent pair;
- (b) $\beta_d(\pi v, \pi w) = \beta_d(v, w)$ for the super S_d -action applied to both arguments.

Proof. The symmetry of $\text{sVect}_{\mathbb{C}}$ on homogeneous vectors is the flip times $(-1)^{|x||y|}$, so a composite of symmetries multiplies the scalars (-1) once for each transposition of two odd letters; this is precisely the sign of the induced permutation of the odd letters, which is multiplicative, and any two decompositions realising the same permutation of the labelled homogeneous tensor factors agree by coherence of the symmetric structure. For (a), moving each w_i leftward past $v_{i+1}, \dots, v_d$ costs exactly $(-1)^{\sum_{i < j} |v_j||w_i|}$ on the support of the product of the $b(v_i, w_i)$. For (b), each factor $b(v_i, w_i)$ forces $|v_i| = |w_i|$ on the support, so every interleaved block $(v_i w_i)$ has even total parity; acting by π on both arguments permutes these even blocks, which costs no sign, and merely reindexes the scalar factors. Terms outside the support vanish on both sides. $\square$

5.2 The copairing and the twisted element

Write $C := \omega(\text{coev}) \in V \otimes V$, and for $D \in V \otimes V$ define $L_D: V \rightarrow V$ by $L_D(x) := (\text{id} \otimes b)(D \otimes x)$, with no reordering, so that $L_D(x) = \sum_i u_i b(w_i, x)$ for $D = \sum_i u_i \otimes w_i$. Applying ω to the snake identity, which holds strictly in $\mathcal{C}_f$ by strand unification, gives $L_C = \text{id}_V$; and since b is nondegenerate, $L_D = L_{D'}$ forces $D = D'$, so this property determines C uniquely. In the coordinates above,

$$C = C_0 + C_1, \quad C_0 = \sum_{i=1}^k e_i \otimes e_i, \quad C_1 = \sum_{i=1}^{2\ell} \xi_i \otimes \eta_i,$$

as follows from the verification $L_{C_0+C_1} = \text{id}$ in Lemma 6.2.

Let $\mathbf{p}: V \rightarrow V$ denote the parity operator, and define the *twisted element*

$$\widehat{C} := (\mathbf{p} \otimes \text{id})C = C_0 - C_1, \quad \text{so that} \quad L_{\widehat{C}} = \mathbf{p}.$$

The twisted element is not a copairing, and free circles must never be evaluated through it. Indeed

$$b(C) = k - 2\ell = \text{sdim } V, \quad b(\widehat{C}) = k + 2\ell = \text{str}(\mathbf{p}),$$

and the two values differ whenever $\ell > 0$. Free circles are carried by multiplicativity and by $f(\bigcirc) = \text{tr}(\text{id}_X) = \text{str}(\text{id}_V) = k - 2\ell$, the middle equality because a symmetric monoidal functor takes categorical traces to supertraces; they are removed *before* the parity transfer of Lemma 5.5.

Both elements are supersymmetric: $c_{V,V}(C) = C$ and $c_{V,V}(\widehat{C}) = \widehat{C}$, since on the odd part $c(\xi_i \otimes \eta_i) = -\eta_i \otimes \xi_i$ while $\sum_i \eta_i \otimes \xi_i = -\sum_i \xi_i \otimes \eta_i$ by the antisymmetry verified in Lemma 6.2(i), and the two signs cancel. This is what licenses assigning the two tensor legs of C , and likewise of $\widehat{C}$, to the two half-edges of an edge without choosing an order.

One further identity involving C is load-bearing enough to merit a name; it is the mechanism by which inserted copairings will fuse back into a single one in Lemma 5.4.

Lemma 5.2 (Copairing fusion). *Write $C = \sum_i u_i \otimes w_i$ with homogeneous terms. Then*

$$\sum_{i,j} (-1)^{|w_i||u_j|} b(w_i, w_j) u_i \otimes u_j = C.$$

Proof. The left-hand side is a fixed linear contraction applied to $C \otimes C$, so it may be computed in the standard presentation $C_0 + C_1$, and the even and odd sectors do not interact since b is even. Even sector: with $u_i = w_i = e_i$ the sign is trivial and $\sum_{i,j} b(e_i, e_j) e_i \otimes e_j = \sum_i e_i \otimes e_i = C_0$. Odd sector: with $u_i = \xi_i$, $w_i = \eta_i$ the sign is -1 throughout, and from the coordinate table of Section 5.1 one computes $b(\eta_i, \eta_j) = \delta_{j,i+\ell}$ for $i \leq \ell$ and $b(\eta_i, \eta_j) = -\delta_{j,i-\ell}$ for $i > \ell$, so

$$-\sum_{i,j} b(\eta_i, \eta_j) \xi_i \otimes \xi_j = -\sum_{i \leq \ell} \xi_i \otimes \xi_{i+\ell} + \sum_{i > \ell} \xi_i \otimes \xi_{i-\ell} = \sum_i \xi_i \otimes \eta_i = C_1. \quad \square$$

5.3 Vertex tensors, mates, and index raising

For each $d \geq 0$ set $T_d := \omega(\sigma_d) \in V^{\otimes d}$, the image of the d -star of Section 3.2. A star has no preferred order on its legs, and this becomes invariance: for every $\pi \in S_d$ we have $P_\pi \circ \sigma_d = \sigma_d$ in $\mathcal{C}_f$, since gluing strands merely relabels open ends (Lemma 3.1(ii)) and every relabelling of the legs of a star is realised by an automorphism of the underlying fragment; applying ω , which takes P_π to the super-permutation action of π , gives $\pi \cdot T_d = T_d$, i.e. $T_d \in (V^{\otimes d})^{S_d}$. Each T_d is moreover *even*, because morphisms from $\mathbf{1}$ are parity-preserving; this evenness is the ultimate origin of the Eulerian condition in Definition 5. Let $\varpi_d: V^{\otimes d} \rightarrow \text{Sym}_s^d V$ be the quotient onto the super symmetric power — here and throughout we use the canonical isomorphism of graded algebras $\text{Sym}_s V \cong \text{Sym } V_0 \otimes \Lambda V_1$, so that the functionals of Definition 5 act on super symmetric powers — and consider the functional $\beta_d(T_d, -)$ on $V^{\otimes d}$. By Lemma 5.1(b) it is S_d -invariant and therefore factors uniquely as

$$\beta_d(T_d, -) = h_d^{\text{cat}} \circ \varpi_d, \quad h_d^{\text{cat}} \in (\text{Sym}_s^d V)^*.$$

No identification of invariants with coinvariants is made, and no factor of $d!$ appears; the functional h_d^{cat} is defined by the factorisation alone. Since $\text{Sym}_s V = \bigoplus_{d \geq 0} \text{Sym}_s^d V$, the family $(h_d^{\text{cat}})_{d \geq 0}$ assembles into a single functional $h^{\text{cat}} \in (\text{Sym}_s V)^*$ — the algebraic dual of a direct sum imposes no finiteness condition — and likewise h^{RS} below; it is these assembled functionals that Definition 5 consumes.

For an even tensor $T \in V^{\otimes d}$, equivalently a morphism $\mathbb{C} \rightarrow V^{\otimes d}$ of $\text{sVect}_{\mathbb{C}}$, define the *mate*

$$T^{\flat} := b^{\otimes d} \circ \text{Interleave}_d \circ (T \otimes \text{id}_{V^{\otimes d}}): V^{\otimes d} \rightarrow \mathbb{C},$$

where Interleave_d is the symmetry rearranging $(v_1 \cdots v_d, w_1 \cdots w_d) \mapsto (v_1 w_1 \cdots v_d w_d)$ and $b^{\otimes d}$ applies b to each adjacent pair. By Lemma 5.1(a),

$$T^{\flat}(w) = \beta_d(T, w); \tag{2}$$

this is the single place where the super signs of the vertex pairing are computed in coordinates.

For a closed graph G without free circles, functionals H_v on $V^{\otimes \deg v}$ and an element $D \in V \otimes V$, define the *tensor network*

$$\text{TN}'_{(H_v), D}(G) := \left(\bigotimes_v H_v \right) \circ \text{Regroup}_G \left(\bigotimes_{e \in E} D \right),$$

where Regroup_G is the symmetry taking the word of D -legs in edge order to the word in vertex–half-edge order; it is a permutation of half-edge–indexed positions, well defined as a composite of braidings by Lemma 5.1. For a family $h = (h_d)$ of functionals on the super symmetric powers we write $\text{TN}_{h, D}(G) := \text{TN}'_{(h_{\deg v} \circ \varpi_{\deg v}), D}(G)$; note that the quotient maps are part of the definition. As written, the definition presupposes auxiliary choices — an order of the edges, an order of the vertices and of the half-edges at each vertex, and an assignment of the two legs of D to the two half-edges of each edge — and the following lemma is what makes the value a function of the unordered graph.

Lemma 5.3. *Suppose that every H_v is an even functional invariant under the super $S_{\deg v}$ -action, and that D is even with $c_{V, V}(D) = D$. Then $\text{TN}'_{(H_v), D}(G)$ is independent of all the auxiliary choices above.*

Proof. Two presentations differ by a composite of four moves: a permutation of the edge factors of $\bigotimes_e D$, which costs no Koszul sign because each factor is an even block

(Lemma 5.1); a swap of the two legs of some factors, which is absorbed by $c_{V,V}(D) = D$; a permutation of the half-edge slots at each vertex, which is absorbed by the invariance of H_v ; and a permutation of the vertex blocks, which is absorbed by naturality of the symmetry, with no sign because the H_v are even. Koszul functoriality (Lemma 5.1) then equates the two signed evaluations. $\square$

Every network appearing below satisfies these hypotheses: the elements C and $\hat{C}$ are even and supersymmetric (Section 5.2); the mates T_d^b are even, since T_d and b are, and S_d -invariant by Lemma 5.1(b) and the invariance of T_d ; and the functionals $h_d \circ \varpi_d$ are even whenever h vanishes in odd exterior degree, and invariant by construction.

Lemma 5.4 (Index raising). *For every closed graph G without free circles,*

$$f(G) = \text{TN}'_{(T_{\deg v}^b), C}(G) = \text{TN}_{h^{\text{cat}}, C}(G).$$

Proof. The second equality is (2) together with the factorisation $\beta_d(T_d, -) = h_d^{\text{cat}} \circ \varpi_d$. For the first, apply ω to the star decomposition (1), giving

$$f(G) = \left(\bigotimes_e b \right) \circ P_G \circ \left(\bigotimes_v T_{\deg v} \right), \quad (3)$$

with P_G now denoting the image of the matching symmetry. Both snake identities hold strictly in $\mathcal{C}_f$ by strand unification, and we insert on each of the $2|E|$ wires of (3) the form

$$\text{id}_V = (b \otimes \text{id}_V) \circ (\text{id}_V \otimes C), \quad \text{i.e.} \quad x = \sum_i b(x, u_i) w_i \quad \text{for } C = \sum_i u_i \otimes w_i,$$

chosen so that the star leg is contracted in first argument position, matching the argument order of β_d . This is the snake identity complementary to the one that characterised $L_C = \text{id}_V$ in Section 5.2; both hold strictly, and no reversal of the odd form is involved. Call u_i the *mate leg* and w_i the *edge leg* of the inserted copairing. Regrouping by naturality and coherence of the symmetry (Lemma 5.1): at each vertex, the inserted evaluations pair the star legs of $T_{\deg v}$ against the mate legs, in the order $b(\text{star leg}, u_i)$, which is by definition the mate $T_{\deg v}^b$ evaluated on the mate legs; and at each edge, the original evaluation contracts the two edge legs arriving from its endpoints, leaving the element

$$\sum_{i,j} (-1)^{|w_i||u_j|} b(w_i, w_j) u_i \otimes u_j = C$$

of $V \otimes V$ feeding the two mate-leg slots — the Koszul sign arises from moving w_i past u_j during the regrouping, and the identity is the Copairing fusion lemma (Lemma 5.2). This is precisely the argument-order supersymmetry sign that a basis-expansion proof must track; here it is carried by naturality of the symmetry, and it is paid exactly once, inside Lemma 5.2. The resulting composite is by definition $\text{TN}'_{(T_{\deg v}^b), C}(G)$, well defined on the unordered graph by Lemma 5.3. We illustrate the insertion, the two named legs, and the fusion in Figure 3. $\square$

5.4 Parity transfer

The functional h^{cat} and the copairing C do not yet match the Regts–Sevenster normalisation, which uses the twisted element. The discrepancy is a sign for each edge of the odd sector, and it cancels against a compensating sign at the vertices, as follows. First, h_d^{cat}

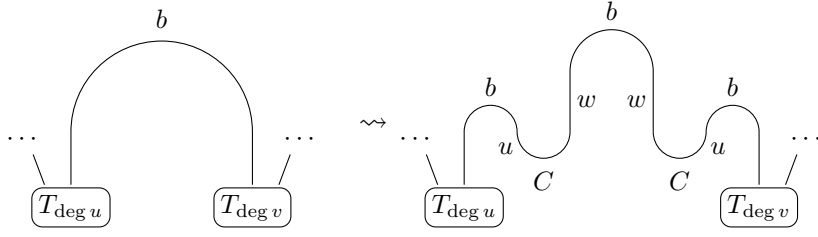


Figure 3: The snake insertion on one edge. The inserted evaluations pair each star leg with the mate leg u of a copairing, assembling the mates T^b ; the edge legs w meet the original evaluation and fuse to a single copairing by Lemma 5.2.

vanishes on $\text{Sym}^a V_0 \otimes \Lambda^j V_1$ for j odd, since T_d is even and β_d is an even pairing. Define the involution J on $\text{Sym} V_0 \otimes \Lambda V_1$ by $J := (-1)^{j(j-1)/2} \cdot \text{id}$ on the summand of exterior degree j — for $j = 2m$ this is $(-1)^m$, and the odd- j values are immaterial by the vanishing just noted — and set

$$h^{\text{RS}} := h^{\text{cat}} \circ J.$$

Lemma 5.5 (Parity transfer). *For every closed graph G without free circles,*

$$\text{TN}_{h^{\text{cat}}, C}(G) = \text{TN}_{h^{\text{RS}}, \hat{C}}(G).$$

Proof. Expand both networks over the set F of edges carrying the odd part of the element (C_1 on the left, $\hat{C}_1$ on the right); the even sectors agree term by term. Fix a surviving term, and let $j_v := \deg_F(v)$ be the number of odd half-edges at v ; terms with some j_v odd vanish on both sides by the odd-degree vanishing above, so every j_v is even. Replacing h^{cat} by h^{RS} multiplies the term by $\prod_v (-1)^{j_v/2} = (-1)^{\frac{1}{2} \sum_v j_v} = (-1)^{|F|}$, by the handshake identity $\sum_v \deg_F(v) = 2|F|$, which is valid for loops precisely because a loop contributes two half-edges. Replacing C_1 by $\hat{C}_1 = -C_1$ multiplies the term by $(-1)^{|F|}$ as well. The two factors cancel. $\square$

A warning about presentations. The involution J is calibrated against the κ -paired form of the vertex argument fixed in Section 2.4. A presentation that instead lists the odd factors in some canonical colour order generates Koszul reordering signs of its own, and the twist must then be recomputed against that presentation rather than carried over unchanged: mixing the two conventions changes every Eulerian summand by exactly $(-1)^{|F|}$ — the sign at stake in Lemma 5.5. The loop example of Section 2.4 serves as the convention check: such a mixture flips its odd sector from -2 to $+2$.

6 The circuit expansion and the Regts–Sevenster formula

Two tasks remain: to prove that the twisted tensor network of Section 5 is *exactly* the Regts–Sevenster sum of Definition 5, circuit signs included; and to assemble the proof of the Main Theorem. The first part is a statement about super linear algebra alone — no connection category enters — and we state it as a theorem in its own right, since it is the step that turns an abstract fibre functor into the concrete model of [29].

Theorem 6.1 (Reconstruction). *Let $V = V_0 \oplus V_1$ be a finite-dimensional super vector space with a nondegenerate supersymmetric even bilinear form b , coordinatised as in Section 5.1,*

let $\widehat{C}$ be the twisted element, and let $h = (h_d)$ be functionals on the super symmetric powers of V vanishing in odd exterior degree. Then for every graph G without free circles,

$$\text{TN}_{h,\widehat{C}}(G) = p_h(G),$$

the right-hand side being the sum of Definition 5 in Section 2.4.

The proof occupies Sections 6.1–6.3. The strategy is to expand the twisted element into its even and odd parts, observe that the even part is diagonal and reproduces the colouring sum over ψ , and reduce the odd part to a sign computation along each circuit; the computation is carried out for a single circuit and shown to be insensitive to everything else.

6.1 The reduction

In the edge order fixed for the network, expand each factor as $\widehat{C} = \widehat{C}_0 + \widehat{C}_1$ and distribute:

$$\bigotimes_{e \in E} \widehat{C} = \sum_{F \subseteq E} \bigotimes_{e \in E} \widehat{C}_{\mathbf{1}_F(e)},$$

where $\mathbf{1}_F$ is the characteristic function of F , so the F -summand carries $\widehat{C}_1$ on the edges of F and $\widehat{C}_0$ elsewhere; both parts being even tensors, the regroupings below introduce no sign on this account. The even part $\widehat{C}_0 = C_0 = \sum_i e_i \otimes e_i$ is diagonal, and its legs contribute, at each vertex, the word of $e_{\psi(a)}$ over the half-edges $a \in \delta_{E \setminus F}(v)$, summed over colourings ψ of the edges of $E \setminus F$; this is precisely the ψ -sum of Definition 5. Since each h_d vanishes in odd exterior degree, a term survives only if every vertex receives an even number of odd half-edges, i.e. only if F is Eulerian. For Eulerian F , choose an Eulerian orientation and a compatible local pairing κ , as in Section 2.4; by [29, Proposition 3] the right-hand side of the Reconstruction Theorem does not depend on these choices. (The dependence is in any case illusory here: the argument below applies to an arbitrary choice, and its conclusion — equality with the manifestly choice-free left-hand side — re-derives this independence for the graphs at hand.) By Lemma 5.3 we may also choose the half-edge order at each vertex so that the even legs occur first and the odd legs occur in κ -paired order; the even and odd legs then combine through the quotient map without interaction: for even vectors $e_{i_1}, \dots, e_{i_a}$ and odd vectors $x_1, y_1, \dots, x_m, y_m$,

$$\varpi_d(e_{i_1} \otimes \dots \otimes e_{i_a} \otimes x_1 \otimes y_1 \otimes \dots \otimes x_m \otimes y_m) = (e_{i_1} \odot \dots \odot e_{i_a}) \otimes (x_1 \wedge y_1) \wedge \dots \wedge (x_m \wedge y_m) \quad (4)$$

under the isomorphism $\text{Sym}_s V \cong \text{Sym } V_0 \otimes \Lambda V_1$: moving even legs past odd legs costs no sign, no factorial normalisation appears, and the right-hand side is exactly the vertex argument of Definition 5. Write $\text{Regroup}_{F,\kappa}$ for the restriction of Regroup_G to the odd legs, targeting the κ -paired vertex–half-edge order. The theorem then reduces to the following identity, applied circuit by circuit.

6.2 The twisted element as maps

Lemma 6.2. *In the coordinates of Section 5.1:*

$$(i) \quad \widehat{C}_1 = -\sum_i \xi_i \otimes \eta_i = \sum_i \eta_i \otimes \xi_i;$$

$$(ii) \quad b(\eta_i, \xi_j) = \delta_{ij} \text{ for all } i, j, \text{ and consequently } L_C = \text{id}_V \text{ and } L_{\widehat{C}} = \mathfrak{p}.$$

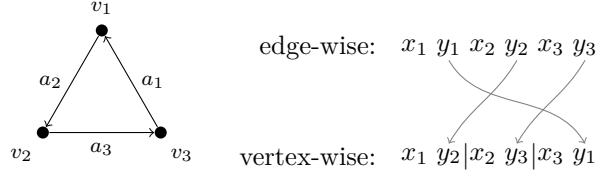


Figure 4: Edge-wise and vertex-wise words on a circuit of length three.

Proof. (i) The first equality is the definition of $\widehat{C} = (\mathfrak{p} \otimes \text{id})C$. For the second, each η_i is a signed ξ , and unwinding the definition gives $\sum_i \eta_i \otimes \xi_i = \sum_{m \leq \ell} (-\xi_{m+\ell} \otimes \xi_m + \xi_m \otimes \xi_{m+\ell}) = -\sum_i \xi_i \otimes \eta_i$. (ii) For $i \leq \ell$: $b(\eta_i, \xi_j) = -b(\xi_{i+\ell}, \xi_j) = -(-\delta_{ij}) = \delta_{ij}$; for $i > \ell$: $b(\eta_i, \xi_j) = b(\xi_{i-\ell}, \xi_j) = \delta_{ij}$, using $b(\xi_m, \xi_{m+\ell}) = 1$. Hence $L_{C_1}(\xi_j) = \sum_i \xi_i b(\eta_i, \xi_j) = \xi_j$, while $L_{C_0}(e_j) = e_j$ directly, and the cross-parity terms vanish because b is even; so $L_C = \text{id}_V$, and $\widehat{C}_1 = -C_1$ gives $L_{\widehat{C}} = \mathfrak{p}$. $\square$

6.3 The circuit sign

Consider a single κ -circuit of length n : edges $a_1, \dots, a_n$ traversed in order through vertices $v_1, \dots, v_n$, the head half-edge x_j of a_j landing in the in-slot of v_j and the tail half-edge y_j in the out-slot of v_{j-1} (indices modulo n).

Lemma 6.3. *The permutation from the edge-wise word $(x_1, y_1, x_2, y_2, \dots, x_n, y_n)$ to the vertex-wise word $(x_1, y_2 | x_2, y_3 | \dots | x_n, y_1)$ fixes the positions $1, 3, \dots, 2n-1$ and acts on the positions $2, 4, \dots, 2n$ as a single n -cycle. All $2n$ legs being odd, its Koszul sign is $(-1)^{n-1}$.*

Proof. The x_j occupy the positions $1, 3, \dots, 2n-1$ in both words, in the same order; the y 's move by one step of a cyclic rotation, which is an n -cycle on the remaining positions, of sign $(-1)^{n-1}$; by Lemma 5.1 the Koszul sign of a rearrangement of an all-odd word is the sign of the permutation. We illustrate the two words in Figure 4. $\square$

Three small auxiliaries dispose of the interactions between circuits. First, the quotient map sends an interleaved product of odd pairs to the wedge — the all-odd case of (4). Second, each pair $x \wedge y$ of odd vectors is an even block, so reordering local pairs at a vertex, and moving complete circuit blocks past one another — including circuits sharing a vertex — introduces no sign. Third, each factor $\widehat{C}_1$ is a sum of terms with two odd legs, hence an even block, so reordering the factors of $\widehat{C}_1^{\otimes |F|}$ into circuit-consecutive order costs no sign; the edge order in Lemma 6.3 may therefore be assumed circuit-consecutive.³

Lemma 6.4. *With notation as above,*

$$\left(\bigotimes_v \varpi_{\deg_F v} \right) \circ \text{Regroup}_{F, \kappa}(\widehat{C}_1^{\otimes |F|}) = (-1)^{c(\kappa)} \sum_{\varphi: F \rightarrow [2\ell]} \bigotimes_v \bigwedge_{(a_1, a_2) \in \kappa_v} \xi_{\varphi(a_1)} \wedge \eta_{\varphi(a_2)}.$$

Proof. By the third auxiliary the edge factors are in circuit-consecutive order, and by the second the identity factors over the circuits of κ ; fix one circuit, of length n . Writing each

³This normalisation is the only point at which the circuit decomposition enters the proof, and it can be avoided: writing W for the walk permutation of the half-edges of F determined by κ and the orientation, the orbits of $W|_{\text{out}}$, its restriction to outgoing half-edges, are the circuits, so $\text{sgn}(W|_{\text{out}}) = (-1)^{|F| - c(\kappa)}$, i.e. $(-1)^{c(\kappa)} = (-1)^{|F|} \text{sgn}(W|_{\text{out}})$; the total regrouping parity is the same permutation sign, and the two combine as in Lemma 6.4 with no decomposition into circuits. We keep the circuitwise proof for its transparency.

of its edge factors as $\widehat{C}_1 = -\sum_i \xi_i \otimes \eta_i$ (Lemma 6.2(i)), with the head leg carrying ξ and the tail leg carrying η — an assignment licensed by the supersymmetry of $\widehat{C}$ (Section 5.2) — the n edge factors contribute the sign $(-1)^n$ in total. The regrouping to the vertex-wise word contributes $(-1)^{n-1}$ (Lemma 6.3). The resulting vertex-wise word is $\bigotimes_j \xi_{\varphi(a_j)} \otimes \eta_{\varphi(a_{j+1})}$, constant colourings and mixed colourings alike being carried along; the quotient maps convert it to the displayed wedges by the first auxiliary, with no further sign. The total sign per circuit is $(-1)^n (-1)^{n-1} = -1$, independent of n , giving $(-1)^{c(\kappa)}$ overall. Loops are the case $n = 1$ and pairs of parallel edges the case $n = 2$; the half-edge conventions of Section 2.1 make both cases instances of the general computation, with nothing further to check. $\square$

Proof of the Reconstruction Theorem. Combine Lemma 6.4 with the reduction of Section 6.1: for each Eulerian F , the odd sector produces the sign $(-1)^{c(\kappa)}$ together with the φ -sum of wedges, the even sector produces the ψ -sum, and applying the functionals $h_{\deg v}$ at the vertices yields exactly the summand of Definition 5 for F . $\square$

The loop example of Section 2.4 keeps its promise here. For the loop graph L , the odd sector of $\text{TN}_{h,\widehat{C}}(L)$ is the case $n = 1$ of Lemma 6.4: the single edge factor $\widehat{C}_1$ contributes $(-1)^1$, the cyclic regrouping is the identity permutation, of sign $(-1)^0$, and the product is the per-circuit -1 ; the φ -sum then passes through the common wedge $\xi_\varphi \wedge \eta_\varphi$ for both odd colours, while the even sector contributes $h(e_\psi \odot e_\psi)$. The network thus reproduces, term by term, the hand computation of Section 2.4.

6.4 Multiplicativity, circles, and the proof of the Main Theorem

Definition 5 is multiplicative over disjoint union: the Eulerian subset, both colourings, and the local circuit data of a disjoint union all split componentwise, and $c(\kappa)$ is additive; the circle convention extends this to graphs with free circles.

Proof of the Main Theorem. The implication from mixed partition functions to exponentially bounded rank is [29, Theorem 6]. For the converse, let f satisfy (H1) and (H2). Write an arbitrary graph as $G = G^\circ \sqcup \bigcirc^r$ with G° free of circles. Then

$$\begin{aligned}
f(G) &= f(G^\circ) f(\bigcirc)^r && \text{(Lemma 3.2)} \\
&= (k - 2\ell)^r \text{TN}_{h^{\text{cat}}, C}(G^\circ) && \text{(Section 5.2, Lemma 5.4)} \\
&= (k - 2\ell)^r \text{TN}_{h^{\text{RS}}, \widehat{C}}(G^\circ) && \text{(Lemma 5.5)} \\
&= (k - 2\ell)^r p_{h^{\text{RS}}}(G^\circ) && \text{(the Reconstruction Theorem)} \\
&= p_{h^{\text{RS}}}(G) && \text{(circle convention),}
\end{aligned}$$

so $f = p_{h^{\text{RS}}}$ is a mixed partition function on $\mathbb{C}^{k|2\ell}$, and $k, 2\ell \leq \lfloor 2eR \rfloor$ by Corollary 4.10. $\square$

7 Consequences and final remarks

We close with an observation, a remark, and a taking of stock.

Observation 7.1 (The ordinary case). *If some super exterior power of X vanishes in $\mathcal{D}_f$ — that is, $\Phi_n(e_{(1^n)}) = 0$ for some $n \geq 1$ — then $V_1 = 0$, and f is an ordinary edge-colouring model.*

Proof. Applying ω as in the proof of Corollary 4.10 gives $\Lambda_s^n V = 0$, where Λ_s denotes the super exterior power. But $\Lambda_s^n V \cong \bigoplus_{i+j=n} \Lambda^i V_0 \otimes \text{Sym}^j V_1$, and if $V_1 \neq 0$ the summand $i = 0, j = n$ is nonzero for every n . Hence $V_1 = 0$, the model has no odd colours, every subset $F \neq \emptyset$ contributes nothing, and p_h reduces to an ordinary edge-colouring partition function. $\square$

In this way the characterisation theorem of [32] for ordinary edge-colouring models sits inside the Main Theorem as the sector in which an exterior power dies; we have not attempted a precise translation between this condition and the real-valuedness hypotheses of [32], which would take us too far afield.

Finally, we take stock. Before the Main Theorem, exponentially bounded edge-connection rank was a necessary condition for being a mixed partition function; it is now the defining one, and the ordinary/skew/mixed family of edge-colouring models closes at the boundary that rank condition draws. The proof has two working parts, and neither is specific to this conjecture: the connection category turns rank conditions on graph parameters into growth conditions on a tensor category, where structural results such as [13] apply; and the reconstruction theorem turns the abstract output of Deligne's theorem back into the concrete normalisations of a combinatorial model. It is the author's hope that both will find further use beyond the theorem proved here.

A Rational trace zeta functions from exponential endomorphism growth

This appendix has two aims: to prove, by a direct symmetric-function argument, that exponential endomorphism growth makes the trace zeta function of every endomorphism rational, with explicit degree bounds; and thereby to give a proof of the nilpotent-trace step of Section 4 that is independent of [13]. Nothing in this appendix depends on Sections 4–6 or on the citations made there; the only inputs are the classical representation theory of symmetric groups and some symmetric function theory, for which we cite [30, 23, 15], and the standard trace calculus of rigid symmetric categories (cyclicity, multiplicativity over $\otimes$, and $\text{tr}(c \circ (u \otimes v)) = \text{tr}(uv)$; see e.g. [12, §4.7]), which for the connection category was proved combinatorially in Lemma 3.5.

A.1 Statement

Throughout the appendix, $\mathcal{C}$ is a rigid symmetric $\mathbb{C}$ -linear monoidal category with $\text{End}(\mathbf{1}) = \mathbb{C}$, and Z is an object of $\mathcal{C}$ satisfying

$$\dim \text{End}(Z^{\otimes n}) \leq A^n \quad (n \geq 0) \quad (5)$$

for a real constant $A \geq 1$. For $g \in \text{End}(Z)$ we write $\tau_m := \text{tr}(g^m)$ and define the *trace zeta function*

$$\zeta_g(z) := \exp\left(\sum_{m \geq 1} \frac{\tau_m}{m} z^m\right) \in \mathbb{C}[[z]].$$

Theorem A.1. *Let s be any integer with $s > 2e\sqrt{A}$. Then for every $g \in \text{End}(Z)$ the trace zeta function is rational,*

$$\zeta_g(z) = \frac{P_g(z)}{Q_g(z)} \quad \text{with} \quad \deg P_g \leq s-1, \quad \deg Q_g \leq s-1,$$

where P_g and Q_g are coprime polynomials with constant term 1. Equivalently, there are disjoint multisets of nonzero complex numbers $\{\alpha_1, \dots, \alpha_{a'}\}$ and $\{\beta_1, \dots, \beta_{b'}\}$ with $a', b' \leq s-1$ and

$$\tau_m = \sum_{i=1}^{a'} \alpha_i^m - \sum_{j=1}^{b'} \beta_j^m \quad (m \geq 1).$$

We call the pair of multisets the *super-spectrum* of g : it is defined as the root data of P_g and Q_g , and we do not claim that its members are eigenvalues of g in any further sense.

Corollary A.2. *Every nilpotent $g \in \text{End}(Z)$ has $\text{tr}(g) = 0$; indeed $\tau_m = 0$ for all $m \geq 1$.*

Etingof and Penneys [13], whose theorem Section 4 uses, work in the broader braided setting and prove a structural theorem: non-negligible objects of moderate growth are rigid, nilpotent quantum traces vanish, and a semisimplification exists. The present result is symmetric and objectwise — it concerns one dualizable object at a time, with no Cauchy completeness, no trace nondegeneracy and no fibre functor entering — and by a direct symmetric-function argument it proves more in that setting: the full super-spectrum of every endomorphism, with degree bounds explicit in the growth constant, and an elementary explanation of *why* nilpotent traces vanish. We have not found the quantitative objectwise formulation of Theorem A.1 in the literature; the closely related prior work is discussed in Section A.5.

The question answered by Corollary A.2 has some history. André and Kahn write, at the opening of [2, §7.3]: “*Nous ignorons si un endomorphisme nilpotent est toujours de trace nilpotente, sans hypothèse supplémentaire.*” Affirmative answers were previously available under Kimura-finiteness [2, Thm 9.2.2, Prop 7.3.3], in the presence of a fibre functor to an abelian rigid target [2, Prop 7.3.3], and now in braided categories of moderate growth [13]. Corollary A.2 gives an independent quantitative proof in the objectwise exponential-growth setting, without invoking a fibre functor, a Kimura decomposition, or the semisimplification machinery.

A.2 Fat-hook confinement

We use the standard representation theory of the symmetric group S_n over $\mathbb{C}$ [30, 15]: the group algebra decomposes into simple blocks $\mathbb{C}[S_n] = \bigoplus_{\lambda \vdash n} B_\lambda$ with $B_\lambda \cong \text{Mat}_{d_\lambda}(\mathbb{C})$, indexed by partitions λ of n with irreducible dimensions d_λ ; the central primitive idempotent of B_λ is

$$e_\lambda = \frac{d_\lambda}{n!} \sum_{\pi \in S_n} \chi_\lambda(\pi) \pi.$$

The symmetric braiding makes S_n act on $Z^{\otimes n}$, giving an algebra homomorphism Φ_n from $\mathbb{C}[S_n]$ to $\text{End}(Z^{\otimes n})$. We say the partition λ *survives* if $\Phi_n(e_\lambda) \neq 0$, and *dies* otherwise. Since $\ker \Phi_n$ is a two-sided ideal, hence a sum of blocks, the whole block B_λ of a surviving λ embeds into $\text{End}(Z^{\otimes n})$; this is the entire mechanism by which (5) acts.

Lemma A.3. *If $\lambda \vdash n$ survives, then $d_\lambda^2 \leq A^n$.*

Proof. The block $B_\lambda \cong \text{Mat}_{d_\lambda}(\mathbb{C})$ embeds into $\text{End}(Z^{\otimes n})$, whose dimension is at most A^n by (5). $\square$

Lemma A.4. *Death is upward closed: if $\lambda \vdash n$ dies and $\lambda \subseteq \mu \vdash n+m$, then μ dies.*

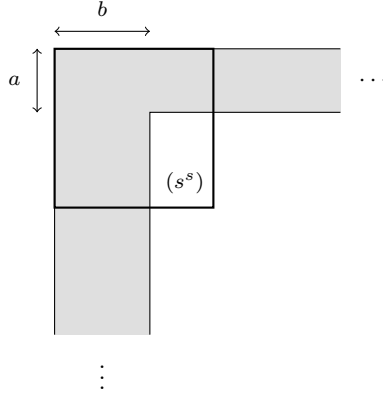


Figure 5: The fat hook $H(a, b)$ and a dead square.

Proof. We first record the branching containment: if $\lambda \subseteq \mu$ then the irreducible V_λ appears in the restriction of V_μ to S_n . Indeed, nested Young diagrams are joined by a chain $\lambda = \nu_0 \subset \nu_1 \subset \dots \subset \nu_m = \mu$ adding one box at a time; the branching rule [30, Thm 2.8.3] gives that V_{ν_i} appears in the restriction of $V_{\nu_{i+1}}$ to the next symmetric group down, and “appears in the restriction” is transitive by semisimplicity.

Consequently the element $e_\mu(e_\lambda \otimes 1_m)e_\mu$ of the simple block $B_\mu \subseteq \mathbb{C}[S_{n+m}]$ is nonzero: it is the compression to V_μ of the projection onto the λ -isotypic part of the restriction, which is a nonzero operator by the previous paragraph. If μ survived, then Φ_{n+m} would be injective on B_μ , so $\Phi_{n+m}(e_\mu(e_\lambda \otimes 1)e_\mu) \neq 0$; but $\Phi_{n+m}(e_\lambda \otimes 1) = \Phi_n(e_\lambda) \otimes \text{id} = 0$ since λ dies. This contradiction proves the lemma. $\square$

Lemma A.5. *Let s be an integer with $s > 2e\sqrt{A}$. Then the square partition (s^s) dies.*

Proof. Every hook length of the square (s^s) is less than $2s$, so the hook length formula and $n! \geq (n/e)^n$ — one term of the series $e^n = \sum_k n^k/k!$ — give, with $n = s^2$,

$$\log d_{(s^s)} \geq \log \frac{(s^2)!}{(2s)^{s^2}} \geq s^2(\log s - 1 - \log 2).$$

The hypothesis $s > 2e\sqrt{A}$ says precisely that $\log s - 1 - \log 2 > \frac{1}{2} \log A$, whence $d_{(s^s)}^2 > A^{s^2}$, and Lemma A.3 kills (s^s) . We have made no attempt to optimise the constant. $\square$

For integers $a, b \geq 0$ let $H(a, b) := \{\lambda : \lambda_{a+1} \leq b\}$ denote the *fat hook*: the set of partitions whose Young diagram fits in the union of a rows and b columns (Figure 5). A partition lies outside $H(a, b)$ if and only if it contains the rectangle $((b+1)^{a+1})$.

Corollary A.6. *With s as in Lemma A.5, every surviving partition lies in the fat hook $H(s-1, s-1)$.*

Proof. A partition outside $H(s-1, s-1)$ contains the square (s^s) , which dies; apply Lemma A.4. $\square$

We now convert confinement into a statement about traces. For a permutation π with cycle lengths $c_1, \dots, c_r$, the standard trace calculus recalled at the head of this appendix gives, by induction on the cycles,

$$\text{tr}(\Phi_n(\pi) \circ g^{\otimes n}) = \prod_{i=1}^r \tau_{c_i}. \quad (6)$$

Indeed, each cycle of π threads the corresponding tensor factors of $g^{\otimes n}$ into a single power of g , closed into a trace. For a partition $\lambda \vdash n$, write $s_\lambda[\tau]$ for the Schur polynomial in the power sums evaluated at $p_m \mapsto \tau_m$; concretely, define $h_0 := 1$, $h_n := 0$ for $n < 0$, and $nh_n := \sum_{i=1}^n \tau_i h_{n-i}$ by Newton's identities, and $s_\lambda[\tau] := \det(h_{\lambda_i - i + j})_{1 \leq i, j \leq r}$, with r the number of parts of λ , by Jacobi–Trudi [23, Chapter I].

Lemma A.7 (Frobenius formula). *For every $\lambda \vdash n$,*

$$\mathrm{tr}(\Phi_n(e_\lambda) \circ g^{\otimes n}) = d_\lambda \cdot s_\lambda[\tau].$$

In particular, if λ dies then $s_\lambda[\tau] = 0$.

Proof. Expand e_λ and apply (6) to each term: writing z_μ for the order of the centraliser of a permutation of cycle type μ , so that the number of such permutations is $n!/z_\mu$,

$$\mathrm{tr}(\Phi_n(e_\lambda) \circ g^{\otimes n}) = \frac{d_\lambda}{n!} \sum_{\pi} \chi_\lambda(\pi) \prod_i \tau_{c_i(\pi)} = d_\lambda \sum_{\mu \vdash n} z_\mu^{-1} \chi_\lambda(\mu) p_\mu[\tau] = d_\lambda s_\lambda[\tau],$$

the last step by the Frobenius characteristic formula $s_\lambda = \sum_{\mu} z_\mu^{-1} \chi_\lambda(\mu) p_\mu$ [23, I.7]. If λ dies, the left-hand side is the trace of the zero morphism composed with $g^{\otimes n}$, and $d_\lambda \neq 0$. $\square$

Combining Corollary A.6 with Lemma A.7: for every $g \in \mathrm{End}(Z)$,

$$s_\lambda[\tau] = 0 \quad \text{for every } \lambda \notin H(s-1, s-1). \quad (7)$$

A.3 Rationality of hook-confined sequences

The passage from (7) to Theorem A.1 is a piece of classical symmetric function theory. The rationality criterion involved goes back to Borel [4]; the formulation through vanishing Hankel determinants is the determinantal rationality of Larsen and Lunts [20, §2], and the assembly via Jacobi–Trudi, in the equivalent language of λ -rings, is due to Mazza and Weibel [25, Prop. 4.14, Cor. 4.15]. We include a self-contained proof, partly to keep the appendix elementary and partly because we want the sharp hook-shaped degree bounds; the argument is a Jacobi–Trudi null-vector computation.

Lemma A.8. *Let $(\tau_m)_{m \geq 1}$ be a complex sequence with $s_\lambda[\tau] = 0$ for every $\lambda \notin H(a, b)$. Then there are coprime polynomials P and Q with constant term 1, $\deg P \leq b$ and $\deg Q \leq a$, such that $H(z) := \sum_{n \geq 0} h_n z^n = P(z)/Q(z)$; consequently, factoring $Q(z) = \prod_i (1 - \alpha_i z)$ and $P(z) = \prod_j (1 - \beta_j z)$, we have $\tau_m = \sum_i \alpha_i^m - \sum_j \beta_j^m$, where the α_i and the β_j form disjoint multisets of nonzero numbers of sizes at most a and b respectively.*

Proof. For strictly decreasing integers $\rho_1 > \dots > \rho_{a+1} \geq b - a$, set $\lambda_i := \rho_i + i$. Then $\lambda_i - \lambda_{i+1} = \rho_i - \rho_{i+1} - 1 \geq 0$, so λ is a partition, and $\lambda_{a+1} = \rho_{a+1} + a + 1 \geq b + 1$, so $\lambda \notin H(a, b)$; the Jacobi–Trudi identity gives

$$\det(h_{\rho_i + j})_{i, j \leq a+1} = s_\lambda[\tau] = 0.$$

Hence the vectors $v_\rho := (h_{\rho+1}, \dots, h_{\rho+a+1})$ for $\rho \geq b - a$ span a space of dimension at most a , and we may choose a nonzero $c = (c_1, \dots, c_{a+1})$ annihilating that span under the standard bilinear pairing: $\sum_{k=1}^{a+1} c_k h_{\rho+k} = 0$ for all $\rho \geq b - a$. Let k^* be maximal with $c_{k^*} \neq 0$ and normalise $c_{k^*} = 1$; the relation reads

$$h_n + q_1 h_{n-1} + \dots + q_{k^*-1} h_{n-k^*+1} = 0 \quad \text{for all } n \geq N := b - a + k^*.$$

Set $Q_0(z) := 1 + q_1 z + \cdots + q_{k^*-1} z^{k^*-1}$, of degree at most $k^* - 1 \leq a$. Then $P_0 := Q_0 H$ has vanishing coefficients in every degree $n \geq N$, since the coefficient of $z^{\rho+k^*}$ in $Q_0 H$ is exactly the left-hand side of the relation at ρ . Note that $N \geq 1$: the relation at $\rho = -k^*$ would read $\sum_k c_k h_{k-k^*} = c_{k^*} h_0 = 1$, every other index being negative, so $\rho = -k^*$ must lie outside the admissible range, i.e. $-k^* < b - a$. Hence $\deg P_0 \leq N - 1 = b - a + k^* - 1 \leq b$, using $k^* \leq a + 1$. Divide P_0 and Q_0 by their greatest common divisor, normalised so that the resulting coprime P and Q retain constant term 1; factoring $Q = \prod_i (1 - \alpha_i z)$ and $P = \prod_j (1 - \beta_j z)$ as in the statement, the logarithmic derivative identity $zH'(z)/H(z) = \sum_{m \geq 1} \tau_m z^m$ — Newton's identities in generating function form — yields $\tau_m = \sum_i \alpha_i^m - \sum_j \beta_j^m$. $\square$

Lemma A.9. *In the situation of Lemma A.8, if $\tau_m = 0$ for all sufficiently large m , then $\tau_m = 0$ for all $m \geq 1$.*

Proof. The series $\Sigma(z) := \sum_{m \geq 1} \tau_m z^m$ has finite support, hence is a polynomial, and $\Sigma = zH'/H = z(P'Q - PQ')/(PQ)$. Thus PQ divides $z(P'Q - PQ')$. Since P divides zPQ' , it divides $zP'Q$; since $\gcd(P, Q) = 1$ and $P(0) = 1$ gives $\gcd(P, zQ) = 1$, we get $P \mid P'$, which in characteristic zero forces $P' = 0$, so P is constant. Symmetrically Q is constant. Hence $H = 1$, every h_n with $n \geq 1$ vanishes, and Newton's identities give $\tau_m = 0$ for all m . $\square$

A.4 Proof of Theorem A.1

Proof. Let $g \in \text{End}(Z)$ and let $s > 2e\sqrt{A}$ be an integer. By (7), the sequence τ satisfies the hypothesis of Lemma A.8 with $a = b = s - 1$, and $H(z)$ is precisely $\zeta_g(z)$: both are the unique power series with constant term 1 whose logarithmic derivative is $\sum_m \tau_m z^{m-1}$. The lemma provides the rational form with its degree bounds and the super-spectrum. $\square$

Proof of Corollary A.2. If g is nilpotent, say $g^M = 0$, then $\tau_m = \text{tr}(g^m) = 0$ for $m \geq M$, and Lemma A.9 extends the vanishing to all $m \geq 1$; in particular $\text{tr}(g) = \tau_1 = 0$. $\square$

A.5 Relation to the literature

The circle of ideas around Theorem A.1 has a substantial history in the theory of motives, which we now locate precisely; the reader interested only in the graph-theoretic results may skip this subsection.

An object Z of a pseudo-abelian rigid symmetric category is *Kimura-finite* if it splits as $Z_+ \oplus Z_-$ with some exterior power of Z_+ and some symmetric power of Z_- vanishing [19, 1], and *Schur-finite* if some Schur functor annihilates it [10, 24]. Kimura-finiteness implies Schur-finiteness [24, Cor. 1.5], and the converse fails: O'Sullivan's example of an indecomposable Schur-finite object that is neither even nor odd appears in [2, 10.1.1] and [24, Ex. 1.12]. Like Kimura-finiteness, our hypothesis (5) forces a fat-hook vanishing pattern (Corollary A.6) — at the level of vanishing patterns exactly the notion of a *bound* in the λ -ring formalism of Mazza and Weibel [25, Def. 4.1] — so the image of Z in the pseudo-abelian envelope of $\mathcal{C}$ is Schur-finite, with a rectangular bound; we claim no implication in either direction between (5) and Kimura-finiteness. Upward closure of the vanishing set, automatic for objects (Lemma A.4; [24, Prop. 1.4(2)]), fails for virtual elements of λ -rings [25, Ex. 4.4], which is why their definition builds the ideal in.

Rationality of zeta functions under such hypotheses is classical in outline. For Kimura-finite objects, rationality (and a functional equation) is due to André, Heinloth and Kahn [1, 17, 18]; for Schur-finite objects of *semisimple* categories it is proved by Kahn [18, Part I]; and the underlying rationality criterion through vanishing Hankel determinants goes back

to Borel [4], in the form given by Larsen and Lunts [20, §2]. Mazza and Weibel assemble exactly this argument in λ -rings and prove their virtual splitting principle for reduced λ -rings [25, Prop. 4.14, Cor. 4.15]; since trace sequences live in the ring of big Witt vectors of $\mathbb{C}$, which is reduced — in characteristic zero its ghost coordinates embed it into a product of copies of $\mathbb{C}$ — the sequence-level content of Lemma A.8 is theirs, and we claim no progress on their virtual splitting conjecture [25, Conj. 4.12], which remains open beyond the reduced case. What Theorem A.1 contributes is the *categorical* statement: the growth bound (5) alone forces the hook confinement (Section A.2) — neither semisimplicity, nor a Kimura decomposition, nor any fibre functor is assumed — and the rationality then follows for every endomorphism, with degree bounds explicit in the growth constant.

The fat hook itself is the territory of Berele and Regev [3], whose hook Schur functions govern the Schur support of super vector spaces (compare [10, Cor. 1.9]). The closest neighbour of our argument in the motivic literature is the work of Del Padrone and Mazza [8, 9], who use the Berele–Regev theory to prove nilpotency of endomorphisms universally of trace zero — the converse direction to Corollary A.2 — under a sign-property hypothesis. Finally, Guletskii [16] shows that rationality of the zeta function does not imply Kimura-finiteness.

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